

Case Study Report

```
library(readxl)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.1
v ggplot2    3.5.2      v tibble     3.2.1
v lubridate  1.9.4      v tidyr      1.3.1
v purrr      1.0.4
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(dplyr)
library(ggplot2)
library(tidyr)
library(knitr)
library(lubridate)
library(naniar)
library(lme4)
```

Loading required package: Matrix

Attaching package: 'Matrix'

The following objects are masked from 'package:tidyr':

expand, pack, unpack

```
forcedeck <- read_excel("data/forcedeck.xlsx")
```

Warning: Expecting numeric in D2925 / R2925C4: got 'Auto'

Warning: Expecting numeric in D2926 / R2926C4: got 'Auto'

Warning: Expecting numeric in D4473 / R4473C4: got 'Auto'

Warning: Expecting numeric in D4474 / R4474C4: got 'Auto'

Warning: Expecting numeric in D4475 / R4475C4: got 'Auto'

```
forceframe <- read_excel("data/forceframe.xlsx")  
nordbord <- read_excel("data/nordbord.xlsx")  
perch <- read_excel("data/perch.xlsx")
```

Warning: Expecting logical in DY5012 / R5012C129: got 'PowerClean'

Warning: Expecting logical in DY5013 / R5013C129: got 'PowerClean'

Warning: Expecting logical in DY5014 / R5014C129: got 'PowerClean'

Warning: Expecting logical in DY5015 / R5015C129: got 'PowerClean'

Warning: Expecting logical in DY5016 / R5016C129: got 'PowerClean'

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Warning: Expecting logical in DY5441 / R5441C129: got 'PowerClean'

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Warning: Expecting logical in DY6092 / R6092C129: got 'PowerClean'

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Warning: Expecting logical in DY7596 / R7596C129: got 'PowerClean'

Warning: Expecting logical in AD7604 / R7604C30: got 'LEFT'

Warning: Expecting logical in AM7604 / R7604C39: got 'SEQUENTIAL'

Warning: Expecting logical in DY7604 / R7604C129: got 'PushPress'

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Warning: Expecting logical in AM11744 / R11744C39: got 'SEQUENTIAL'

Warning: Expecting logical in AD11745 / R11745C30: got 'LEFT'

Warning: Expecting logical in AM11745 / R11745C39: got 'SEQUENTIAL'

Warning: Expecting logical in AD11746 / R11746C30: got 'LEFT'

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Warning: Expecting logical in DY19790 / R19790C129: got 'PowerClean'

Warning: Expecting logical in DY19791 / R19791C129: got 'PowerClean'

Warning: Expecting logical in DY19792 / R19792C129: got 'PowerClean'

Warning: Expecting logical in DY19977 / R19977C129: got 'PowerClean'

Warning: Expecting logical in DY19978 / R19978C129: got 'PowerClean'

Warning: Expecting logical in DY19979 / R19979C129: got 'PowerClean'

Warning: Expecting logical in DY19980 / R19980C129: got 'PowerClean'

Warning: Expecting logical in DY19981 / R19981C129: got 'PowerClean'

Warning: Expecting logical in DY19990 / R19990C129: got 'PowerClean'

Warning: Expecting logical in DY19991 / R19991C129: got 'PowerClean'

Warning: Expecting logical in DY19992 / R19992C129: got 'PowerClean'

Warning: Expecting logical in DY19993 / R19993C129: got 'PowerClean'

Warning: Expecting logical in DY19994 / R19994C129: got 'PowerClean'

Warning: Expecting logical in DY20216 / R20216C129: got 'PowerClean'

Warning: Expecting logical in DY20217 / R20217C129: got 'PowerClean'

Warning: Expecting logical in DY20218 / R20218C129: got 'PowerClean'

Warning: Expecting logical in DY20219 / R20219C129: got 'PowerClean'

Warning: Expecting logical in DY20220 / R20220C129: got 'PowerClean'

Warning: Expecting logical in DY20221 / R20221C129: got 'PowerClean'

Warning: Expecting logical in DY20222 / R20222C129: got 'PowerClean'

Warning: Expecting logical in DY20603 / R20603C129: got 'PowerClean'

Warning: Expecting logical in DY20604 / R20604C129: got 'PowerClean'

Warning: Expecting logical in DY20605 / R20605C129: got 'PowerClean'

Warning: Expecting logical in DY20606 / R20606C129: got 'PowerClean'

Warning: Expecting logical in DY20607 / R20607C129: got 'PowerClean'

Warning: Expecting logical in DY20608 / R20608C129: got 'PowerClean'

Warning: Expecting logical in DY20609 / R20609C129: got 'PowerClean'

Warning: Expecting logical in DY20617 / R20617C129: got 'PowerClean'

Warning: Expecting logical in DY20618 / R20618C129: got 'PowerClean'

Warning: Expecting logical in DY20619 / R20619C129: got 'PowerClean'

Warning: Expecting logical in DY20620 / R20620C129: got 'PowerClean'

Warning: Expecting logical in DY20844 / R20844C129: got 'PowerClean'

Warning: Expecting logical in DY20845 / R20845C129: got 'PowerClean'

Warning: Expecting logical in DY20846 / R20846C129: got 'PowerClean'

Warning: Expecting logical in DY20847 / R20847C129: got 'PowerClean'

Warning: Expecting logical in DY20848 / R20848C129: got 'PowerClean'

Warning: Expecting logical in DY20849 / R20849C129: got 'PowerClean'

Warning: Expecting logical in DY20850 / R20850C129: got 'PowerClean'

Warning: Expecting logical in DY20860 / R20860C129: got 'PowerClean'

Warning: Expecting logical in DY20861 / R20861C129: got 'PowerClean'

Warning: Expecting logical in DY20862 / R20862C129: got 'PowerClean'

Warning: Expecting logical in DY20863 / R20863C129: got 'PowerClean'

Warning: Expecting logical in DY20864 / R20864C129: got 'PowerClean'

Warning: Expecting logical in DY20891 / R20891C129: got 'CleanPull'

Warning: Expecting logical in DY20892 / R20892C129: got 'CleanPull'

Warning: Expecting logical in DY20893 / R20893C129: got 'CleanPull'

Warning: Expecting logical in DY20894 / R20894C129: got 'CleanPull'

Warning: Expecting logical in DY21108 / R21108C129: got 'PowerClean'

Warning: Expecting logical in DY21109 / R21109C129: got 'PowerClean'

Warning: Expecting logical in DY21110 / R21110C129: got 'PowerClean'

Warning: Expecting logical in DY21111 / R21111C129: got 'PowerClean'

Warning: Expecting logical in DY21112 / R21112C129: got 'PowerClean'

Warning: Expecting logical in DY21113 / R21113C129: got 'PowerClean'

Warning: Expecting logical in DY21114 / R21114C129: got 'PowerClean'

Warning: Expecting logical in DY21123 / R21123C129: got 'PowerClean'

Warning: Expecting logical in DY21124 / R21124C129: got 'PowerClean'

Warning: Expecting logical in DY21125 / R21125C129: got 'PowerClean'

Warning: Expecting logical in DY21126 / R21126C129: got 'PowerClean'

Warning: Expecting logical in DY21127 / R21127C129: got 'PowerClean'

Warning: Expecting logical in DY21365 / R21365C129: got 'PowerClean'

Warning: Expecting logical in DY21366 / R21366C129: got 'PowerClean'

Warning: Expecting logical in DY21367 / R21367C129: got 'PowerClean'

Warning: Expecting logical in DY21368 / R21368C129: got 'PowerClean'

Warning: Expecting logical in DY21369 / R21369C129: got 'PowerClean'

Warning: Expecting logical in DY21370 / R21370C129: got 'PowerClean'

Warning: Expecting logical in DY21371 / R21371C129: got 'PowerClean'

Warning: Expecting logical in DY21372 / R21372C129: got 'PowerClean'

Warning: Expecting logical in DY21381 / R21381C129: got 'PowerClean'

Warning: Expecting logical in DY21382 / R21382C129: got 'PowerClean'

Warning: Expecting logical in DY21383 / R21383C129: got 'PowerClean'

Warning: Expecting logical in DY21384 / R21384C129: got 'PowerClean'

Warning: Expecting logical in DY21385 / R21385C129: got 'PowerClean'

Warning: Expecting logical in DY21631 / R21631C129: got 'PowerClean'

Warning: Expecting logical in DY21632 / R21632C129: got 'PowerClean'

Warning: Expecting logical in DY21633 / R21633C129: got 'PowerClean'

Warning: Expecting logical in DY21634 / R21634C129: got 'PowerClean'

Warning: Expecting logical in DY21635 / R21635C129: got 'PowerClean'

Warning: Expecting logical in DY21636 / R21636C129: got 'PowerClean'

Warning: Expecting logical in DY21637 / R21637C129: got 'PowerClean'

Warning: Expecting logical in DY21646 / R21646C129: got 'PowerClean'

Warning: Expecting logical in DY21647 / R21647C129: got 'PowerClean'

Warning: Expecting logical in DY21648 / R21648C129: got 'PowerClean'

Warning: Expecting logical in DY21649 / R21649C129: got 'PowerClean'

Warning: Expecting logical in DY21650 / R21650C129: got 'PowerClean'

Warning: Expecting logical in DY21728 / R21728C129: got 'PowerClean'

Warning: Expecting logical in DY21729 / R21729C129: got 'PowerClean'

Warning: Expecting logical in DY21730 / R21730C129: got 'PowerClean'

Warning: Expecting logical in DY21731 / R21731C129: got 'PowerClean'

Warning: Expecting logical in DY21732 / R21732C129: got 'PowerClean'

Warning: Expecting logical in DY21733 / R21733C129: got 'PowerClean'

Warning: Expecting logical in DY22100 / R22100C129: got 'PowerClean'

Warning: Expecting logical in DY22101 / R22101C129: got 'PowerClean'

Warning: Expecting logical in DY22102 / R22102C129: got 'PowerClean'

Warning: Expecting logical in DY22103 / R22103C129: got 'PowerClean'

Warning: Expecting logical in DY22104 / R22104C129: got 'PowerClean'

Warning: Expecting logical in DY22105 / R22105C129: got 'PowerClean'

Warning: Expecting logical in DY22106 / R22106C129: got 'PowerClean'

Warning: Expecting logical in DY22115 / R22115C129: got 'PowerClean'

Warning: Expecting logical in DY22116 / R22116C129: got 'PowerClean'

Warning: Expecting logical in DY22117 / R22117C129: got 'PowerClean'

Warning: Expecting logical in DY22118 / R22118C129: got 'PowerClean'

Warning: Expecting logical in DY22362 / R22362C129: got 'PowerClean'

Warning: Expecting logical in DY22363 / R22363C129: got 'PowerClean'

Warning: Expecting logical in DY22364 / R22364C129: got 'PowerClean'

Warning: Expecting logical in DY22365 / R22365C129: got 'PowerClean'

Warning: Expecting logical in DY22366 / R22366C129: got 'PowerClean'

Warning: Expecting logical in DY22367 / R22367C129: got 'PowerClean'

Warning: Expecting logical in DY22368 / R22368C129: got 'PowerClean'

Warning: Expecting logical in DY22376 / R22376C129: got 'PowerClean'

Warning: Expecting logical in DY22377 / R22377C129: got 'PowerClean'

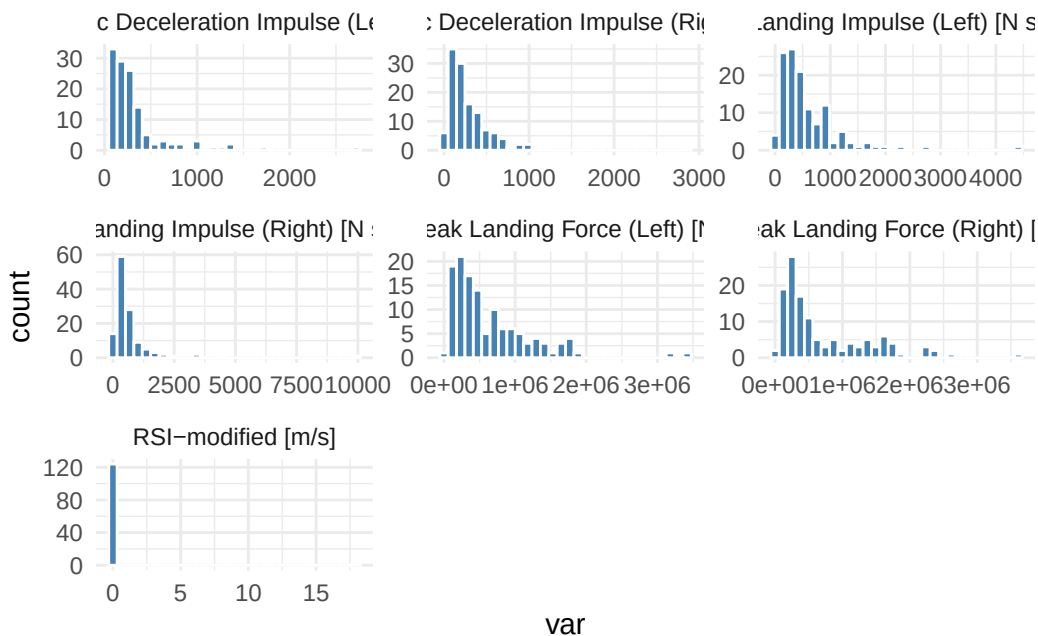
Warning: Expecting logical in DY22378 / R22378C129: got 'PowerClean'

Warning: Expecting logical in DY22379 / R22379C129: got 'PowerClean'

```
# ---- forcedeck ----
forcedeck_vars <- c(
  "Athlete Id",
  "RSI-modified [m/s]",
  "Peak Landing Force (Left) [N]",
  "Peak Landing Force (Right) [N]",
  "Landing Impulse (Left) [N s]",
  "Landing Impulse (Right) [N s]",
  "Eccentric Deceleration Impulse (Left) [N s]",
  "Eccentric Deceleration Impulse (Right) [N s]"
)

forcedeck_long <- forcedeck |>
  select(all_of(forcedeck_vars)) |>
  pivot_longer(~`Athlete Id`, names_to = "variable", values_to = "value") |>
  group_by(`Athlete Id`, variable) |>
  summarise(var = var(value, na.rm = TRUE), .groups = "drop")

ggplot(forcedeck_long, aes(x = var)) +
  geom_histogram(bins = 30, fill = "steelblue", color = "white") +
  facet_wrap(~variable, scales = "free") +
  theme_minimal()
```

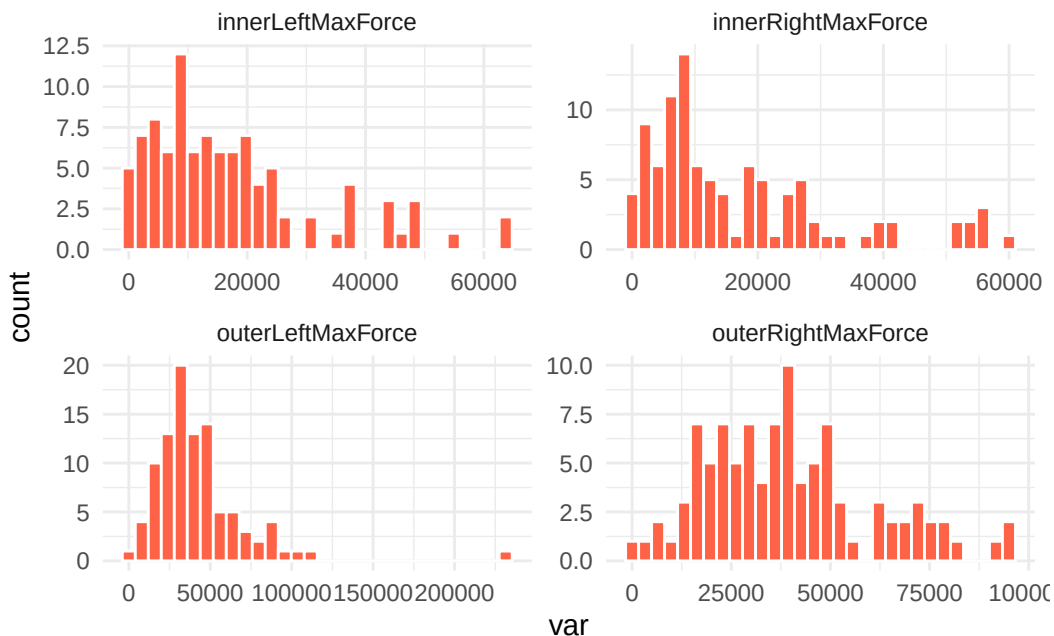


```
# ---- forceframe ----
forceframe_vars <- c(
  "athleteId",
  "innerRightMaxForce",
  "innerLeftMaxForce",
  "outerRightMaxForce",
  "outerLeftMaxForce"
)

forceframe_long <- forceframe |>
  select(all_of(forceframe_vars)) |>
  pivot_longer(-athleteId, names_to = "variable", values_to = "value") |>
  group_by(athleteId, variable) |>
  summarise(var = var(value, na.rm = TRUE), .groups = "drop")

ggplot(forceframe_long, aes(x = var)) +
  geom_histogram(bins = 30, fill = "tomato", color = "white") +
  facet_wrap(~variable, scales = "free") +
  theme_minimal()
```

Warning: Removed 36 rows containing non-finite outside the scale range (`stat\_bin()`).

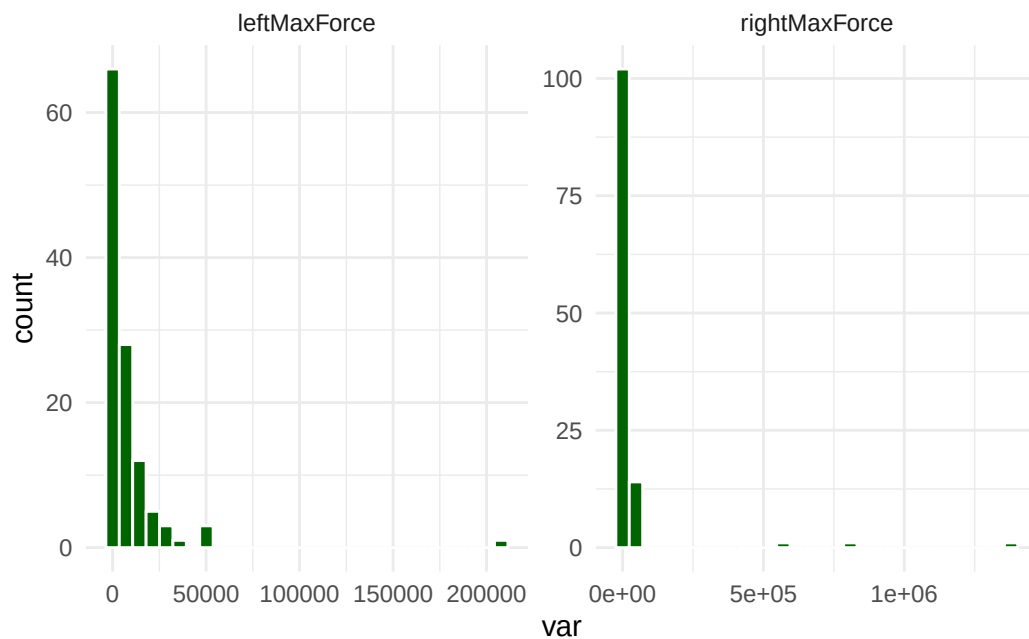


```
# ---- nordbord ----
nordbord_vars <- c(
  "athleteId",
  "rightMaxForce",
  "leftMaxForce"
)

nordbord_long <- nordbord |>
  select(all_of(nordbord_vars)) |>
  pivot_longer(-athleteId, names_to = "variable", values_to = "value") |>
  group_by(athleteId, variable) |>
  summarise(var = var(value, na.rm = TRUE), .groups = "drop")

ggplot(nordbord_long, aes(x = var)) +
  geom_histogram(bins = 30, fill = "darkgreen", color = "white") +
  facet_wrap(~variable, scales = "free") +
  theme_minimal()
```

Warning: Removed 8 rows containing non-finite outside the scale range (`stat\_bin()`).



```
# ---- perch ----
perch_vars <- c(
  "name",
  "max_peak_power"
)

perch_long <- perch |>
  rename(`Max Peak Power` = max_peak_power) |>
  select(name, `Max Peak Power`) |>
  pivot_longer(-name, names_to = "variable", values_to = "value") |>
  group_by(name, variable) |>
  summarise(var = var(value, na.rm = TRUE), .groups = "drop")

ggplot(perch_long, aes(x = var)) +
  geom_histogram(bins = 30, fill = "purple", color = "white") +
  facet_wrap(~variable, scales = "free") +
  theme_minimal()
```

